UUM
Universiti Utara Malaysia

FINAL EXAMINATION
FIRST SEMESTER 2024/2025 SESSION

COURSE CODE/NAME	:	STIKK2114 / OPERATING SYSTEM
DATE	:	22 JANUARY 2025 (WEDNESDAY)
TIME	:	9.00 A.M. – 11.30 A.M. (2 ½ HOURS)
VENUE	:	BK8 COLGIS

INSTRUCTIONS :

1. This question paper has **THREE (3) SECTIONS: A, B,** and **C**, in **ELEVEN (11)** printed pages excluding the cover page. **SECTION A** has **TWENTY (20)** questions, **SECTION B** has **EIGHT (8)** questions and **SECTION C** has **NINE (9)** questions.
2. Answer **ALL** questions in **SECTION A** on the answer sheet on page 3. Answer **ALL** questions in **SECTION B** and **C** in the space provided.
3. You are **NOT ALLOWED** to remove the question paper from the examination hall.
4. Candidates are subject to the **UUM ACADEMIC DISHONESTY REGULATION PROCEDURES.**

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS QUESTION PAPER UNTIL INSTRUCTED

SECTION A: SHORT ANSWER QUESTIONS (20 MARKS)

INSTRUCTIONS: Fill in the answer space provided on page 3 with the correct answer based on the options provided in the box below:

asymmetric	parity	starvation	address	implicit
sniffer	race	authenticate	thread	structure
pits	busy waiting	traffic	indivisible	lands
encapsulation	block	high-level	compression	path
lower-level	resources	independent	symmetric	livelock
adaptability	spoofing	explicit	survivability	improved

1. Job Scheduler is the _____ scheduler of the Processor Manager that selects jobs from a queue of incoming jobs based on each job's characteristics.

2. Aging is a policy used to ensure that jobs that have been in the system for a long time in the _____ queues will eventually complete their execution.

3. Process Control Block (PCB) is a data _____ that contains information about the current status and characteristics of a process.

4. Safe state is the situation in which the system has enough available _____ to guarantee the completion of at least one job running on the system.

5. _____ is similar to a deadlock except that neither process is blocked or obviously waiting; both are in a continuous state of change.

6. While deadlock affects systemwide performance, _____ affects individual jobs or processes.

7. _____ parallelism is a type of concurrent programming in which the compiler automatically detects which instructions can be performed in parallel.

8. Test-and-set (TS) is an _____ machine instruction, which is executed in a single machine cycle to determine whether the processor is available.

9. Master/slave configuration is a/an _____ multiprocessing configuration consisting of a single processor system connected to "slave" processors each of which is managed by the primary "master" processor, which provides the scheduling functions and jobs.

10. _____ is a term used to describe the state in which processes, waiting for an event to occur, continuously test to see if the condition has changed and remain in unproductive, resource consuming wait loops.

11. The modules of the I/O subsystem that monitor the status of every device, control unit, and channel are known as I/O _____ controller.

12. _____ is a term used for tiny depressions on the reflective layer of an optical disc.

13. _____ bit is an extra bit added to a character, word, or other data unit and used for error checking.

14. The industry has amended the RAID acronym to mean Redundant Array of _____ Disks to emphasize the scheme's improved disk performance and reliability.

15. In a direct file organization environment, relative _____ indicates the position of a record relative to the beginning of the file.

16. A procedure used to reduce the amount of space required to store data by reducing, encoding, or abbreviating repetitive terms or characters is known as data _____.

17. _____ is the sequence of directories and subdirectories the operating system must follow to find a specific file.

18. Picture password is a sequence of strokes over a picture or graphic that is used to _____ access to a computer system by an authorized user.

19. System _____ is defined as the capability of a system to fulfill its mission, in a timely manner, in the presence of attacks, failures, or accidents.

20. The creation of false IP addresses in the headers of data packets sent over the Internet, sometimes with the intent of gaining access when it would not otherwise be granted is known as _____.

ANSWER FOR SECTION A

1.		11.	
2.		12.	
3.		13.	
4.		14.	
5.		15.	
6.		16.	
7.		17.	
8.		18.	
9.		19.	
10.		20.	

SECTION B: STRUCTURED QUESTIONS (30 MARKS)
INSTRUCTIONS: Answer **ALL** the questions in the space provided.

1. State whether the following process scheduling algorithm is **PREEMPTIVE** or **NON-PREEMPTIVE**.

Process Scheduling Algorithm	Preemptive / Non-preemptive
Priority Scheduling	
Earliest Deadline First	
Shortest Remaining Time	

(3 marks)

2. Match each of the following descriptions with one of four conditions for deadlock.

Descriptions	Condition for Deadlock
A process is allowed to hold on to resources while it is waiting for other resources to finish execution.	No preemption
Each process involved is waiting for a resource being held by another; each process is blocked and can't continue, resulting in deadlock.	
Only one process is allowed to have access to a resource.	
Each process refuses to relinquish the resources it holds until its execution is completed even though it isn't using them because it's waiting for other resources.	

(3 marks)

3. What is the only difference between a loosely coupled multiprocessing system and a collection of independent single-processing systems?

(2 marks)

4. How is the increment operation performed by the semaphore to prevent the occurrence of race conditions?

(2 marks)

5. When the topic of process synchronization is discussed; the term critical region is often used.

   a) What is meant by critical region?

(2 marks)

   b) What will happen to the processes within the critical region?

(2 marks)

6. List and explain briefly **THREE (3)** categories of the system's peripheral devices.

(6 marks)

7. Some systems shorten the access control list in more detail by establishing **FOUR (4)** categories of users. List **ALL** user categories.

(4 marks)

8. An alternative to password is the use of biometric to identify a subject.

   a) In your own words, describe what is a biometric.

(1 mark)

   b) List **THREE (3)** examples of biometric systems.

(3 marks)

   c) Passwords and biometrics are said to be unable to prevent Intentional unauthorized access such as Trapdoor. What is meant by Trapdoors?

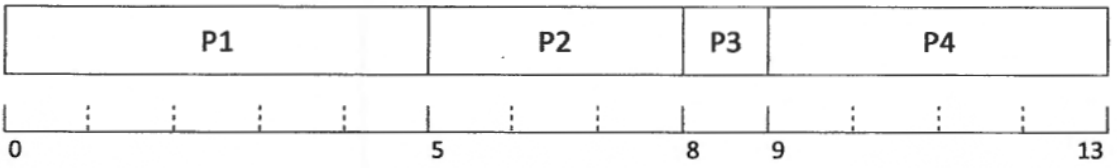
(2 marks)

SECTION C: PROBLEM / COMPUTATIONAL QUESTIONS (50 MARKS)
INSTRUCTIONS: Answer **ALL** the questions in the space provided.

1. Evaluate the diagram below. The diagram simulates a scheduling operation of four processes. Given that whenever apply, the quantum time is 2-unit times.

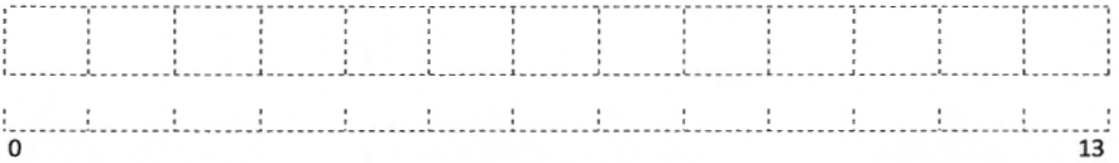
Process	Arrival time	Burst Time
P1	0	5
P2	1	3
P3	2	1
P4	4	4

The chart below shows the sequence of processes and the execution timeline of those processes using the First Come First Serve (FCFS) scheduling algorithm.



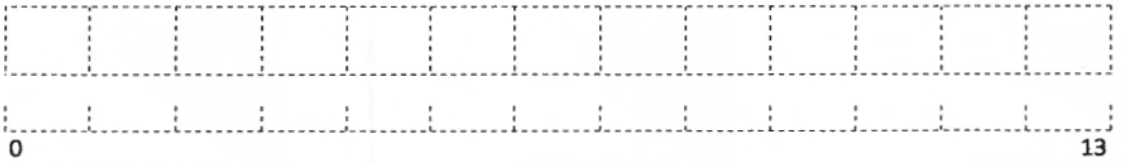
Based on the evaluation you have made, and an example chart as given above, complete a chart (with the sequence of processes and the execution timeline) that represents the following scheduling algorithms:

a. Shortest Job Next (SJN)



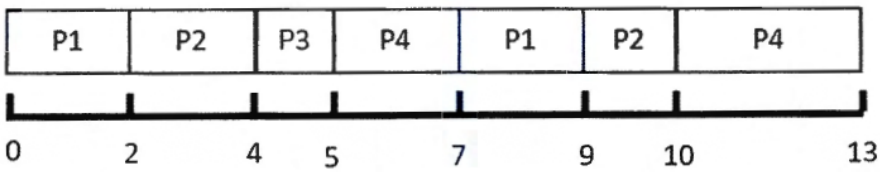
(3 marks)

b. Shortest Remaining Time (SRT)



(3 marks)

2. Based on the following chart, calculate the correct average turnaround time for each process (in the order of P1, P2. P3, P4).



(4 marks)

P1:

P2:

P3:

P4:

3. Given that it takes **2.0 milliseconds** (ms) to travel from one track to the next of a hard drive; that the arm is originally positioned at **Track 15** moving toward the **low-numbered tracks**; and that you are using the **LOOK scheduling policy**. Compute the total seek time to satisfy the following requests — 04, 40, 35, 11, 14, and 07. Assume all requests are initially present in the wait queue. (Ignore rotational time and transfer time; just consider seek time.)

Head Path	Track Travel	Seek Time (ms)
→		
→		
→		
→		
→		
→		
TOTAL SEEK TIME		

(10 marks)

4. Compare the two types of gaps, Inter-record gap (IRG) and Inter-block gap (IBG) found on magnetic tapes used as Sequential Access Storage Media. State **ONE (1)** similarity and **ONE (1)** main difference between these gaps.

(2 marks)

5. There are 3 ways files or records are physically organized by the operating system, namely Sequential, direct and indexed sequential. Compare these three forms of file/record organization. State **ONE (1)** advantage and **ONE (1)** disadvantage of each.

File/record organization	Advantage	Disadvantage
Sequential		
Direct		
Indexed sequential		

(6 marks)

6. The current byte address (CBA) must be updated every time a record is accessed. Briefly discuss how CBA is obtained in **THREE (3)** files/records access methods. Also state the CBA arithmetic expression of each.

a) Sequential Access (fixed-length records)

(3 marks)

b) Sequential Access (variable-length records)

(3 marks)

c) Direct Access (fixed-length records)

(3 marks)

7. **Scenario 1:** A user receives a phone call from a person who claims to represent IT services and then asks that user for confirmation of username and password for auditing purposes.
Scenario 2: A cleaner attempts to enter a computer lab but is denied entry by the receptionist because there is no scheduled cleaning for that day.

a) Based on the above 2 scenarios, which security threat (or technique) does these scenarios represent?

(1 mark)

b) Name the terms used specifically for scenario 1 and scenario 2.

(2 marks)

Scenario 1:

Scenario 2:

c) What does the intruder want or seek from both scenarios?

(2 marks)

8. Identify the correct Security Terms that match with the following descriptions.

Assets	Vulnerability	Threat	Exploit	Mitigation

Descriptions	Security Terms
The counter-measure that reduces the likelihood or severity of a potential threat or risk.	
A weakness in a system, or its design, that could be exploited by a threat.	
A potential danger to a company's assets, data, or network functionality.	
Anything of value to the organization. It includes people, equipment, resources, and data.	

(4 marks)

9. Passwords are one of the easiest and most effective protection schemes to implement in the operating system. Prove that a password created using **SIX (6) Alphanumeric** characters is more secure than **NINE (9) Numeric** digits.

(4 marks)

END OF QUESTIONS